S1. Version, scope, and reproduction

This appendix documents the learned-frontend experiment: 36 clips, 56 sparse endpoints and 7,530 causal frame steps. Office contributes 22 clips/31 endpoints; Cup Games contributes 14/25. They are two TCOW collections, not independent benchmark families or a reserved test. The earlier classical study and its appendix are preserved in archive/v1. The new main manuscript and v2 implementation are authoritative for this run.

Executed methods. Frozen DINOv2 ViT-B/14 appearance, appearance plus single motion, appearance plus a spatial mixture, and the mixture with controlled auxiliary-memory updates. All methods receive the same initial query mask and dense feature proposals. Local flow is not semantic containment. Fixed gates and variance are uncalibrated.

Labels. Endpoint target masks are accessed after predictions are generated. The evaluator does not measure continuous identity switches, first visible reappearance, exact hidden coordinates or real robot task success. Processing full videos does not fill those annotation gaps.

Reproduce. Install the supplied requirements and use the official DINOv2 source repository and ViT-B/14 checkpoint. Paths are arguments; no credentials are required. DINOv2's own dependencies are also required by its hub entry point.

```
python v2/test_model.py
python v2/evaluate.py --data data/rubric_office data/rubric_cup_games --repo /path/to/dinov2 --output v2/results --
    device mps
```

CPU can be selected with –device cpu. This can change runtime. Per-clip traces record the chosen candidate, acceptance, cosine, association gap, age, mixture alternatives and weights. Endpoints, source hashes and selection-change counts are separate JSON records. The local run took approximately 459 seconds for all variants together.

Artifacts. v2/protocol.json records the pre-run configuration; v2/model.py implements inference; v2/evaluate.py owns label access; v2/references_verified.json contains live primary-source metadata and source hashes for 47 references. Model/source hashes are recorded in v2/provenance.json.

Integration. The official Ameca Gen 1 Animator project was created and idle tracks saved. No responsive speech or robot control is validated. v2/speech.py formats evidence and accepts an injected speech callable; it is not a fabricated Tritium API.

S2. Reading the complete case audit

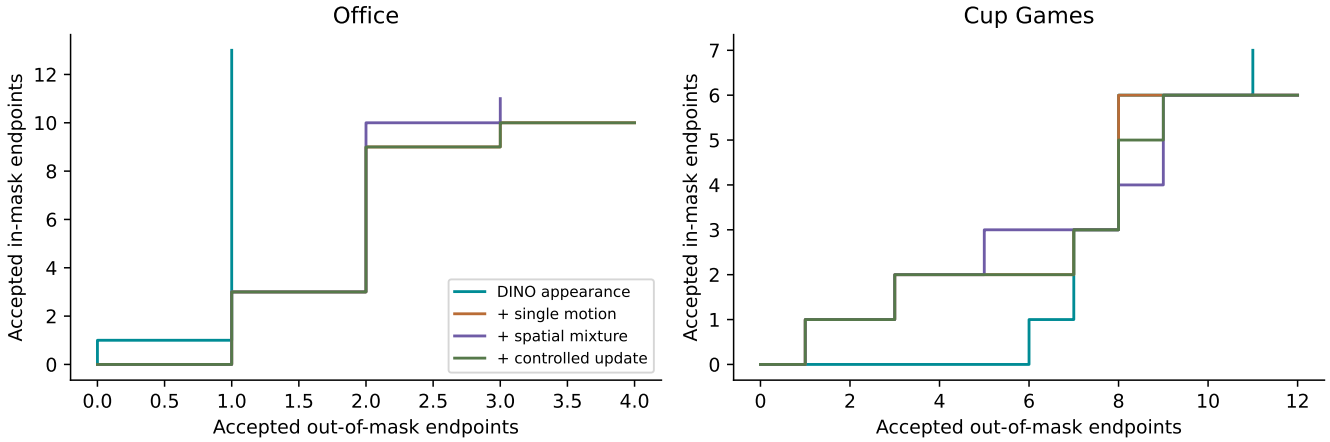
Each following page contains two clips. Every eligible clip is included, sorted by collection and filename. The pictures show the initialization frame and the final scored endpoint under appearance and spatial-mixture selection. Earlier endpoints are retained in the table, even when not pictured. A cyan circle is a predicted candidate, never an occluder or a modified input image.

Table codes. H: accepted and center inside target mask. M: accepted and center outside mask. R+: rejected despite an in-mask center. R-: rejected with an out-of-mask center. A rejection is not a correct identity answer. Columns A/V/M/U denote appearance, single velocity, spatial mixture, and controlled update.

Caution about interpretation. The masks do not establish a dense, physical identity trajectory. Candidate membership is narrower than reacquisition. Data have been inspected and cannot support held-out novelty claims. Endpoints within clips and scenes are dependent.

What the review exposed. The classical confirmation rule changed no evaluated decisions. The integrated mixture now changes 1,934 selections, but increases accepted endpoint errors. That mechanism is active without being beneficial. The new learned appearance control is stronger than the old handcrafted proposal on Office, but it is not a reproduction of SAMURAI, DAM4SAM, Cutie or another full tracking system.

Required next evidence. Independent scenes, dense visibility/identity labels, a relation inference mechanism, current learned tracking baselines with matched supervision, calibration and an executed observation or answering task remain necessary. No reviewer score is inferred from the presence of references or figures.



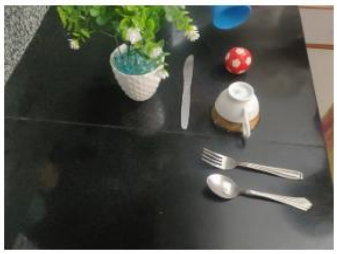
Full outcome accounting. The paper reports every operating point from the same fixed scores. These descriptive curves should not be used to choose test thresholds. Scene-level calibration and evaluation remain a separate unexecuted requirement.

S3. Full-sequence execution records

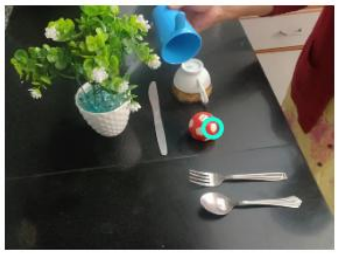
rubric_cup_games / cgt_frames_0002

Query frame 100; processed 170 subsequent frames; pictured endpoint 270.

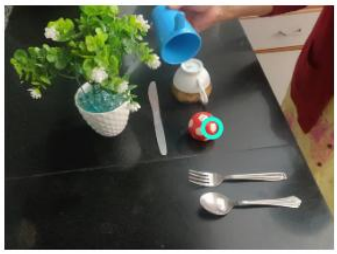
Initial query



DINO appearance



Spatial mixture

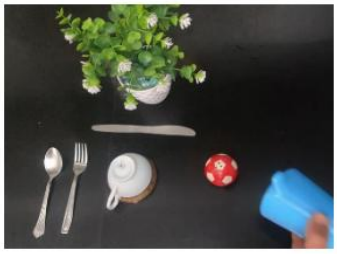


Frame	A	V	M	U
270	H	H	H	H

rubric_cup_games / cgt_frames_0011

Query frame 90; processed 210 subsequent frames; pictured endpoint 300.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
300	H	H	H	H

S4. Full-sequence execution records

rubric_cup_games / cgt_frames_0026

Query frame 60; processed 300 subsequent frames; pictured endpoint 360.

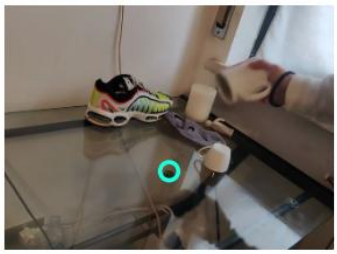
Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
360	H	H	H	H

rubric_cup_games / cgt_frames_0061

Query frame 40; processed 300 subsequent frames; pictured endpoint 340.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
70	R-	R-	R-	R-
140	R-	R-	R-	R-
180	R-	R-	R-	R-
208	R-	R-	R-	R-
235	R-	R-	R-	R-
270	R-	R-	R-	R-
340	M	M	M	M

S5. Full-sequence execution records

rubric_cup_games / cgt_frames_0065

Query frame 80; processed 230 subsequent frames; pictured endpoint 310.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
110	R+	R+	R+	R+
210	M	M	M	M
310	M	M	M	M

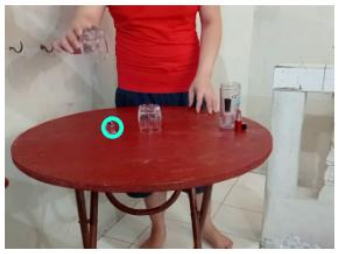
rubric_cup_games / cgt_frames_0076

Query frame 105; processed 213 subsequent frames; pictured endpoint 318.

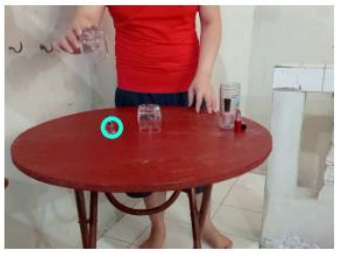
Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
150	M	M	M	M
285	M	M	M	M
318	H	H	H	H

S6. Full-sequence execution records

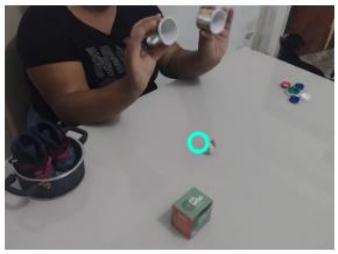
rubric_cup_games / cgt_frames_0092

Query frame 80; processed 330 subsequent frames; pictured endpoint 410.

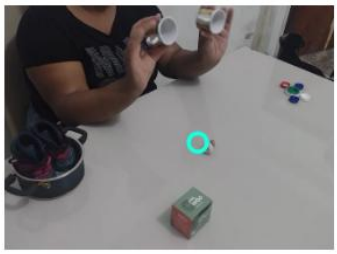
Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
410	M	M	M	M

rubric_cup_games / cgt_frames_0113

Query frame 40; processed 270 subsequent frames; pictured endpoint 310.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
310	M	M	M	M

S7. Full-sequence execution records

rubric_cup_games / cgt_frames_0126

Query frame 80; processed 270 subsequent frames; pictured endpoint 350.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
350	M	M	M	M

rubric_cup_games / cgt_frames_0136

Query frame 195; processed 117 subsequent frames; pictured endpoint 312.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
312	H	H	H	H

S8. Full-sequence execution records

rubric_cup_games / cgt_frames_0137

Query frame 60; processed 240 subsequent frames; pictured endpoint 300.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
300	M	M	M	M

rubric_cup_games / cgt_frames_0154

Query frame 60; processed 255 subsequent frames; pictured endpoint 315.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
315	M	M	M	M

S9. Full-sequence execution records

rubric_cup_games / cgt_frames_0172

Query frame 20; processed 268 subsequent frames; pictured endpoint 288.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
155	R+	M	M	M
288	M	M	M	M

rubric_cup_games / cgt_frames_0211

Query frame 90; processed 250 subsequent frames; pictured endpoint 340.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
340	H	H	H	H

S10. Full-sequence execution records

rubric_office / bagduck1_move

Query frame 20; processed 210 subsequent frames; pictured endpoint 230.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
230	R-	R-	R-	R-

rubric_office / chips1_oof

Query frame 40; processed 240 subsequent frames; pictured endpoint 280.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
160	H	H	H	M
280	R+	M	M	M

S11. Full-sequence execution records

rubric_office / confuse1

Query frame 20; processed 116 subsequent frames; pictured endpoint 136.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
136	H	H	H	H

rubric_office / confuse3

Query frame 15; processed 58 subsequent frames; pictured endpoint 73.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
73	R+	R-	R+	R+

S12. Full-sequence execution records

rubric_office / handball1_wave

Query frame 5; processed 132 subsequent frames; pictured endpoint 137.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
137	M	M	M	M

rubric_office / handball2_boxoccl

Query frame 30; processed 180 subsequent frames; pictured endpoint 210.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
210	H	H	H	H

S13. Full-sequence execution records

rubric_office / handball3_occlpick

Query frame 20; processed 170 subsequent frames; pictured endpoint 190.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
190	H	H	H	H

rubric_office / lightball4_occl1x

Query frame 40; processed 300 subsequent frames; pictured endpoint 340.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
190	R+	R+	R+	R+
312	R-	R-	R-	R-
340	R-	R-	R-	R-

S14. Full-sequence execution records

rubric_office / lightball5_occl2x

Query frame 30; processed 206 subsequent frames; pictured endpoint 236.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
170	R+	R-	R-	R-
236	R-	R-	R-	R-

rubric_office / lightball6_occl4x

Query frame 15; processed 275 subsequent frames; pictured endpoint 290.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
175	R-	R-	R-	R-
226	R-	R-	R-	R-
290	R+	R+	R+	R+

S15. Full-sequence execution records

rubric_office / mugduck2_reveal

Query frame 20; processed 180 subsequent frames; pictured endpoint 200.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
200	R+	R+	R+	R+

rubric_office / mugduck3_reveal

Query frame 15; processed 180 subsequent frames; pictured endpoint 195.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
195	R+	R+	R+	R+

S16. Full-sequence execution records

rubric_office / mugduck5_stay

Query frame 20; processed 120 subsequent frames; pictured endpoint 140.



Frame	A	V	M	U
140	R-	R+	R-	R-

rubric_office / mugduck6_shuffle

Query frame 30; processed 180 subsequent frames; pictured endpoint 210.



Frame	A	V	M	U
210	R-	R-	R-	R-

S17. Full-sequence execution records

rubric_office / mugduck7_shuffle

Query frame 30; processed 180 subsequent frames; pictured endpoint 210.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
190	R+	R+	R+	R+
210	R+	R+	R+	R+

rubric_office / plantcupball2

Query frame 25; processed 240 subsequent frames; pictured endpoint 265.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
265	R+	R+	R+	R+

S18. Full-sequence execution records

rubric_office / pump1_scan

Query frame 30; processed 240 subsequent frames; pictured endpoint 270.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
150	H	R-	H	H
270	R+	H	H	H

rubric_office / pumpcookie1_reveal

Query frame 20; processed 180 subsequent frames; pictured endpoint 200.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
200	H	H	H	H

S19. Full-sequence execution records

rubric_office / teaduck1

Query frame 50; processed 120 subsequent frames; pictured endpoint 170.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
170	H	H	H	H

rubric_office / teaduck2

Query frame 15; processed 180 subsequent frames; pictured endpoint 195.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
195	H	H	H	H

S20. Full-sequence execution records

rubric_office / teaduck3_reveal

Query frame 15; processed 180 subsequent frames; pictured endpoint 195.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
195	H	H	H	H

rubric_office / teaduck4_reveal

Query frame 15; processed 240 subsequent frames; pictured endpoint 255.

Initial query



DINO appearance



Spatial mixture



Frame	A	V	M	U
160	H	H	H	H
255	H	M	M	M